Year 9 Algebra Revision — Basics

Straightforward practice of each skill on its own — no multi-step problems yet.

1 Simplifying Expressions

Adding and subtracting like terms

1. $3x + 5x$ _____
2. $9y - 4y$ _____
3. $4a + 3a - 2a$ _____
4. $5x + 2y + 2x - y$ _____

Multiplying and dividing

5. $3 \times 4x$ _____
6. $(-2) \times 5a$ _____
7. $12x \div 4$ _____
8. $20y \div 5$ _____

2 Solving Equations

One-step equations

1. $x + 7 = 12$ _____
2. $x - 5 = 9$ _____
3. $4x = 24$ _____

Two-step equations

4. $2x + 3 = 11$ _____
5. $3x - 4 = 11$ _____
6. $\frac{x}{2} + 1 = 5$ _____

Multi-step equations

7. $2(x + 3) = 16$ _____
8. $3x + 2x - 1 = 14$ _____
9. $5x - 3 - x = 9$ _____

Variables on both sides

10. $3x + 2 = x + 10$ _____
11. $5x - 1 = 2x + 8$ _____
12. $4x + 3 = 2x + 11$ _____

3 Tables of Values

Complete each table using the given rule.

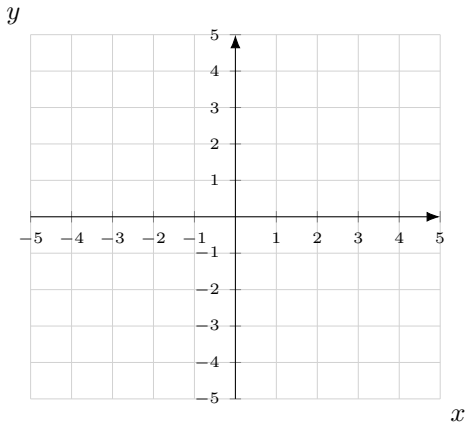
$y = x + 3$						
x	-2	-1	0	1	2	3
y						

$y = 2x - 1$						
x	-2	-1	0	1	2	3
y						

4 Plotting Points

Plot and label each of these points on the Cartesian plane below.

$$(2, 3) \quad (-3, 1) \quad (0, -4) \quad (-2, -2) \quad (4, 0) \quad (-4, 4) \quad (1, -3)$$

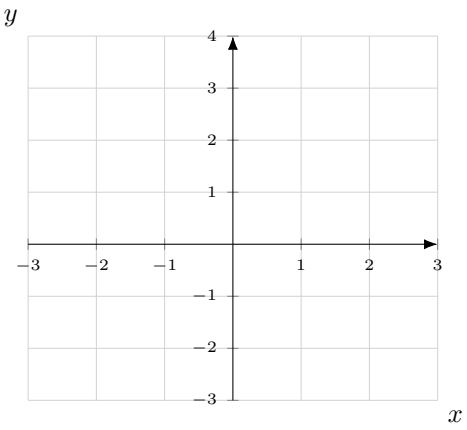


5 Drawing a Line from an Equation

For each equation: complete the table, plot the points on the grid, then draw a straight line through them.

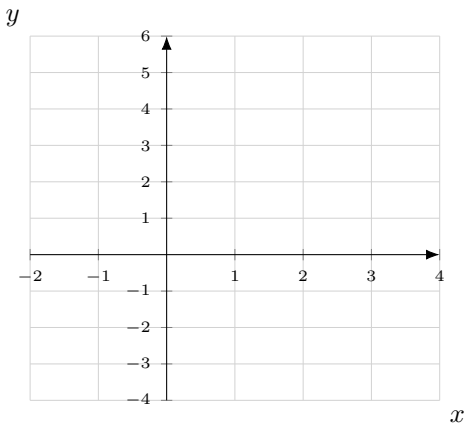
$y = x + 1$

x	-2	-1	0	1	2
y					



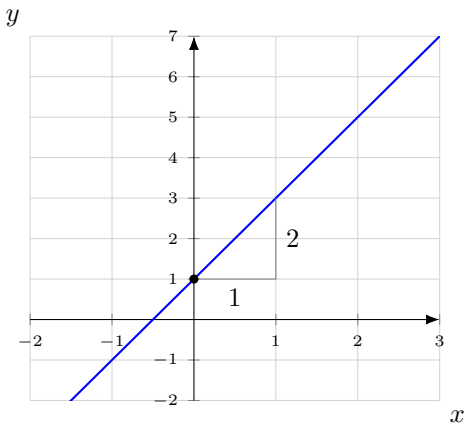
$y = -2x + 3$

x	-1	0	1	2	3
y					



6 Reading a Line: y-intercept, Gradient, Equation

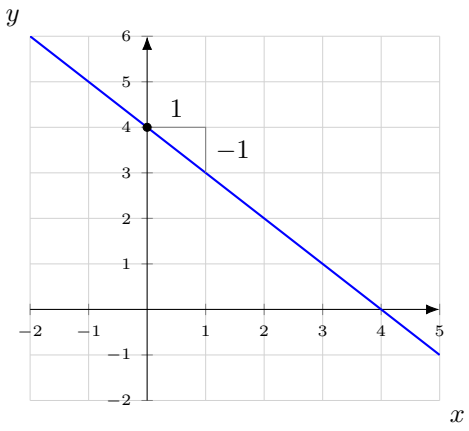
For each graph, find: (a) the y-intercept, (b) the gradient (using the triangle shown), and (c) the equation of the line in the form $y = mx + c$.



(a) y-intercept: _____

(b) Gradient: _____

(c) Equation: _____



(a) y-intercept: _____

(b) Gradient: _____

(c) Equation: _____

7 Reading a Table: y-intercept, Gradient, Equation

For each table, find: (a) the y-intercept, (b) the gradient, and (c) the equation of the line in the form $y = mx + c$.

x	0	1	2	3
y	2	5	8	11

(a) y-intercept: _____

(b) Gradient: _____

(c) Equation: _____

x	0	1	2	3
y	5	3	1	-1

(a) y-intercept: _____

(b) Gradient: _____

(c) Equation: _____